

***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

CS/EC/IT 19-200-0104B & CE/EE/ME/SE 19-200-0204A

BASIC CIVIL ENGINEERING

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(8 × 3 = 24)

- I. Write notes on:
- (a) Field tests for bricks.
 - (b) P.C.C and R.C.C
 - (c) Machine foundation.
 - (d) Compare Stone and Brick masonry.
 - (e) Indirect ranging.
 - (f) Temporary adjustments of dumpy level.
 - (g) IGBC
 - (h) Camber and Super elevation.

PART B

(4 × 12 = 48)

- II. Discuss the different varieties of cement available for construction.
- OR**
- III. List out the different point to be observed while concreting the roof slab of a multi-storeyed building.
- IV. Show the arrangements of bricks for constructing a two-brick thick wall building using English bond for two alternate layers of a corner, also show the elevation for at least 4 layers.
- OR**
- V. Suggest a suitable steel sloping roof for a workshop of 7 meter span, draw the elevation of that truss showing all connections.
- VI. Suppose you want to purchase a land in your locality and before the documentation of the same you want to check the total area, step by step explain the procedure for measuring the area by listing out the obstruction you may come across while doing the work at the site.
- OR**
- VII. The following consecutive readings were taken with a dumpy level.
- 2.500, 2.340, 1.940, 1.850, 0.500, 2.780, 1.650, 0.540, 2.850, 0.450, 0.555**
- The level was shifted after the fifth and eighth readings. The first reading was taken on a bench mark of reduced level **120. 250**. Calculate the reduced levels of all the points then carryout the check.
- VIII. National Highway 66, is a 1,608 km long busy National Highway that runs roughly north-south along the western coast of India, parallel to the Western Ghats, Show its cross section.
- OR**
- IX. The UN reports that, since 2015, over 600 million people have gained access to safely-managed drinking water. Globally, three out of four people had safe drinking water in 2022. In that time, water-use efficiency has increased 10% globally. In this context explain the different sources of water.